

Ahmet Eren Yalvaç

Sabancı University – Industrial Engineering

1. Motivation

Football has always been one of my biggest interests, especially the English Premier League. Since Premier League teams are located in cities with very different economic and social conditions, I became curious about whether those differences could have any effect on football performance.

Some cities are wealthier, have higher salaries, and better living conditions than others. I wanted to explore whether teams from these cities perform better at home compared to teams from less affluent cities. Even though football is mainly influenced by sporting factors such as player quality and tactics, I thought it would be interesting to see if there was also a measurable connection between city-level wealth and match outcomes.

The main question of this project is:

Do wealth-related variables influence the probability of a home team winning in the English Premier League?

2. Data Source

This project combines football match data with socioeconomic data from UK cities.

The football data covers Premier League seasons from 2010–2011 to 2024–2025 and was collected mainly from:

- **Kaggle**
- **Public football-data archives**

The match dataset includes:

- **Match results**
- **Home and away goals**
- **Team information**
- **Match statistics**

To support the analysis, I also collected city-level socioeconomic data related to:

- **Income**
- **Cost of living**

- **Housing affordability**
- **Rent-to-income ratios**

These datasets were merged by matching football clubs with their corresponding cities.

The data collection process involved downloading datasets, cleaning inconsistent values, and combining multiple sources into one final dataset suitable for analysis.

3. Data Analysis

The project was completed in several stages.

Data Cleaning and Preparation

Before analysis, the datasets needed preprocessing. This included:

- **Cleaning missing values**
- **Standardizing formats**
- **Merging football and socioeconomic datasets**
- **Creating new variables related to wealth and affordability**

Variables such as:

- **real_wealth**
- **affordability_ratio**
- **rent_to_income_pct**

were used to represent the economic conditions of cities.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was used to better understand the dataset and identify patterns.

Several visualizations were created, including:

- **Home win distributions**
- **Seasonal trends**
- **Scatter plots**
- **Histograms**

- **Boxplots**

These graphs helped compare socioeconomic variables with football performance and observe whether any visible patterns existed.

Statistical Analysis

Because the target variable (home_win) is binary, I used point-biserial correlation instead of Pearson correlation.

This method is more appropriate when analyzing relationships between:

- **A binary variable**
- **Continuous variables**

The analysis tested whether variables such as wealth and affordability were related to the probability of winning at home.

Machine Learning Models

To further analyze the data, I applied machine learning methods.

Logistic Regression

Logistic Regression was used as a basic classification model to predict whether the home team wins.

Random Forest

A Random Forest model was also implemented to capture more complex relationships between variables.

The models were evaluated using:

- **Accuracy**
- **Confusion Matrix**
- **Classification Report**

4. Findings

The results show that socioeconomic variables alone are not strong predictors of football match outcomes.

The statistical analysis showed:

- **Weak correlations between wealth variables and home wins**

- No strong statistical significance in most tests

The machine learning models also achieved limited predictive accuracy.

This suggests that city-level wealth does not have a major impact on whether a team wins at home.

Instead, football results are likely influenced much more by factors such as:

- Team quality
- Player performance
- Tactics
- Injuries
- Squad depth
- Current form

Overall, the project suggests that football success is much more connected to sporting factors than to the economic conditions of cities.

5. Limitations and Future Work

One limitation of this project is that it mainly focuses on socioeconomic variables and does not include many football-specific performance metrics.

For example, future studies could include:

- Team market value
- Player statistics
- Injury data
- Expected goals (xG)
- ELO ratings

Another limitation is that football outcomes are naturally unpredictable and influenced by many short-term factors that are difficult to measure.

In the future, more advanced machine learning models and additional leagues could also be explored.

6. Conclusion

This project investigated whether socioeconomic conditions influence football match outcomes in the English Premier League.

The findings suggest that:

- **Wealth-related variables have weak relationships with home wins**
- **Machine learning models cannot strongly predict match outcomes using socioeconomic variables alone**

Final Conclusion

City-level socioeconomic conditions do not meaningfully determine football success in the Premier League.

Football outcomes are primarily shaped by team and player-related factors rather than broader economic conditions.